The Perceived Monetary Policy Rule: Identification and Macroeconomic Implications

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Motivation

*"Providing forecasts to illustrate the Fed's **reaction function** works better in theory than in practice, better in the lab than in the field."*

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What is a reaction function, and why does perception about it matter?

Research Question

1. **Reaction function:** how interest rates get adjusted to changing economic conditions.

The Taylor Rule: $\underbrace{i_t}_{\text{Interest rate}} = \underbrace{\phi_{\pi,t}}_{\text{Rule}} \underbrace{\pi_t}_{\text{Inflation}}$

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   - "Is the Fed hawkish?" (Actual Rule ($\phi_{\pi,t}$)) > "Do people *perceive* the Fed as hawkish?" (Perceived Rule ($E_t[\phi_{\pi,t}] \equiv \phi_{\pi,t|t}$))

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Research Question

1. *How can we identify the perception?*
2. *What are the macroeconomic implications of the perception?*

Overview - Identification

1. **Identification of the perceived rule**
   - *Convention:* Run Taylor-rule regressions using interest rate forecasts.

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   - *Taylor principle:* Raise i more than 1:1 with π.
   - *Nowcast-based:* Taylor principle ✓ ($\phi_{\pi,t|t} > 1$).
   - *Forecast-based:* Taylor principle ✗ ($\phi_{\pi,t|t} < 1$).

Overview - Macroeconomic Implications

Subjective Perception Markov Switching New Keynesian (SP-MSNK) model.

Perception ($\phi_{\pi,t|t}$) vs. Actual Rule ($\phi_{\pi,t}$)

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2. **Perception and monetary policy implications**
   - Effective policy requires managing and precisely measuring **perception**.
   - *Great Moderation:* without the shift in **perception**, inflation volatility ×1.72.
   - *Post-2020 inflation surge:* Mismeasured **perception** suggests a sub-optimal lift-off.

Contribution to the literature

1. Macroeconomic effects of policy perceptions: Bauer, Pflueger, and Sunderam 2024, Schorfheide 2005, Hamilton, Pruitt, and Borger 2011, Bocola et al. 2024. . .
   ⇒ Evaluate GE consequences when perceived rules differ from actual rules.
2. Estimating perceived monetary policy rules: Hamilton, Pruitt, and Borger 2011, Bundick 2015, Jia, Shen, and Zheng 2023, Bocola et al. 2024, Bauer, Pflueger, and Sunderam 2024. . .
   ⇒ Identify structural covariance bias and propose nowcast-based identification.
3. Determinacy and the Great Moderation: Lubik and Schorfheide 2004, Coibion and Gorodnichenko 2011, Bianchi 2013, Sims and Zha 2006. . .
   ⇒ Reinterpret the Great Moderation as driven by "Good Perceived Policy".
4. Regime-switching DSGE models: Davig and Leeper 2007, Bianchi 2013, Barthélemy and Marx 2017, Bianchi, Lettau, and Ludvigson 2022. . .
   ⇒ Relax the FIRE assumption to isolate the perception channel.
5. Post-pandemic policy and credibility: Nakamura, Riblier, and Steinsson 2025, Bocola et al. 2025. . .
   ⇒ Show that the perceived Taylor principle expands the central bank's flexibility.

Roadmap

1. **Simple Model of Perceived vs. Actual Rule**
2. Identification of Perceived Rule
3. Empirical Estimation of Perceived Rule
4. Subjective Perception Markov Switching New Keynesian Model

A Simple Model: Why the Perceived Taylor Rule Matters

Setup:

$$i_t = E_t^P[\pi_{t+1}] + r_t, \qquad r_t = \rho_r r_{t-1} + v_t$$

$$i_t = \phi_p \pi_t \quad \text{(perceived)}$$

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$$i_t = \mathbb{E}_t[\pi_{t+1}] + r_t$$

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$$\Rightarrow \pi_t = \frac{1}{\phi_f - \rho_r} r_t \qquad \text{(requires } \phi_f > 1\text{)}$$

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Perceived law of motion (requires $\phi_p > 1$):

$$i_{t|t} \equiv E_t^P[i_t] = \frac{\phi_p}{\phi_p - \rho_r} r_t$$
$$\pi_{t|t} \equiv E_t^P[\pi_t] = \frac{1}{\phi_p - \rho_r} r_t$$

Nowcast ($h = 0$): anticipated realization of the agent.

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Nowcast ($h = 0$): anticipated realization of the agent.

$$E_t^P[i_{t+h}] = \phi_p \cdot \frac{\rho_r^h}{\phi_p - \rho_r} r_t$$
$$E_t^P[\pi_{t+h}] = \frac{\rho_r^h}{\phi_p - \rho_r} r_t$$

A Simple Model: Takeaways

1. **Perception channel** (through $E_t^P[\pi_{t+1}]$) vs. **direct policy channel** (through $1/\phi_f$):

$$\pi_t = \frac{1}{\phi_f}\Bigg(\underbrace{\frac{\rho_r}{\phi_p - \rho_r} r_t}_{E_t^P[\pi_{t+1}]} + r_t\Bigg)$$

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3. **Identification:** regressing $E_t^P[i_{t+h}]$ on $E_t^P[\pi_{t+h}]$ recovers ϕ_p, at any h.

$$\frac{E_t^P[i_{t+h}]}{E_t^P[\pi_{t+h}]} = \frac{\phi_p \cdot \cancel{\frac{\rho_r^h}{\phi_p - \rho_r} r_t}}{\cancel{\frac{\rho_r^h}{\phi_p - \rho_r} r_t}} = \phi_p \qquad \text{for all } h$$

Identification of Perceived Rule

Conventional practice regresses interest rate *forecasts* on inflation *forecasts*:
Hamilton, Pruitt, and Borger (2011) and Bauer, Pflueger, and Sunderam (2024)

$$E_t[i_{t+h}] = \beta_{OLS,IV}\, E_t[\pi_{t+h}] + e_{th} \qquad \Rightarrow \qquad \widehat{\beta}_{OLS,IV} = \boldsymbol{\phi_p}$$

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But the monetary policy rule could move over time:
Kim and Nelson (2006), Coibion and Gorodnichenko (2011), and Bianchi (2013)

$$i_t = \phi_{\pi,t}\pi_t \;\Rightarrow\; E_t[i_{t+h}] = E_t[\phi_{\pi,t+h}]\, E_t[\pi_{t+h}] + \overbrace{\mathrm{Cov}_t(\pi_{t+h}, \phi_{\pi,t+h})}^{\text{OVB}}$$

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$$\mathrm{Cov}\Big(E_t[\pi_{t+h}],\ \mathrm{Cov}_t(\pi_{t+h}, \phi_{\pi,t+h})\Big) < 0 \quad \text{since} \quad \begin{cases} E_t[\pi_{t+h}] > 0, & \mathrm{Cov}_t(\cdot) < 0 \\ E_t[\pi_{t+h}] < 0, & \mathrm{Cov}_t(\cdot) > 0 \end{cases}$$

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The goal: Interest rate forecast provides a biased estimate of **perception**.

Source of Covariance Bias - Illustration

Lubik and Schorfheide (2004) and Farmer, Waggoner, and Zha (2011):

$$
\begin{aligned}
x_t &= \mathbb{E}_t x_{t+1} - \tau \left(i_t - \mathbb{E}_t \pi_{t+1} \right) + z_{D,t}, \\
\pi_t &= \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + z_{S,t}, \\
i_t &= \phi_{\pi, s_t} \pi_t .
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where $z_{D,t}$ and $z_{S,t}$ follow persistent AR(1) processes.

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\phi_{\pi, s_t} \in \{\phi_{\pi,1}, \phi_{\pi,2}\}, \phi_{\pi,1} > \phi_{\pi,2} \text{ with } P = \begin{bmatrix} p & 1-p \\ 1-p & p \end{bmatrix}
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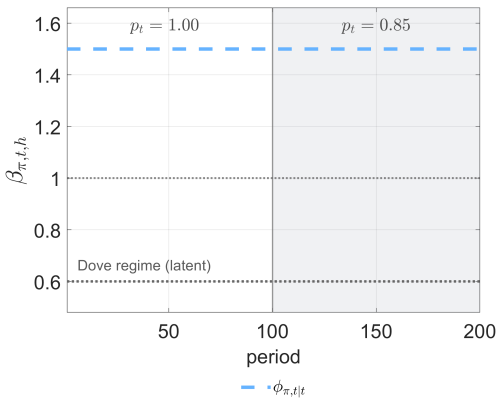
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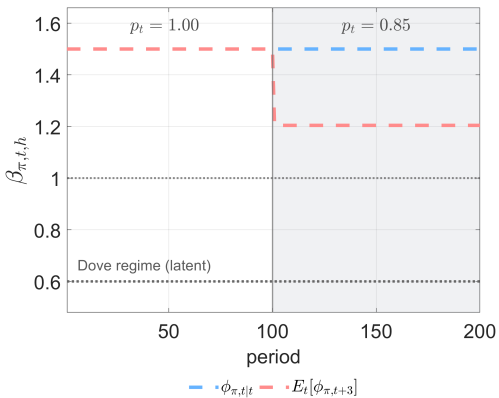
Suppose a forecaster uses this model to report forecasts.

Numerical Illustration - Forecast vs. Nowcast



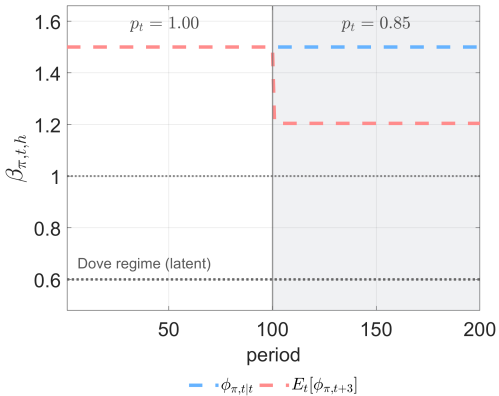
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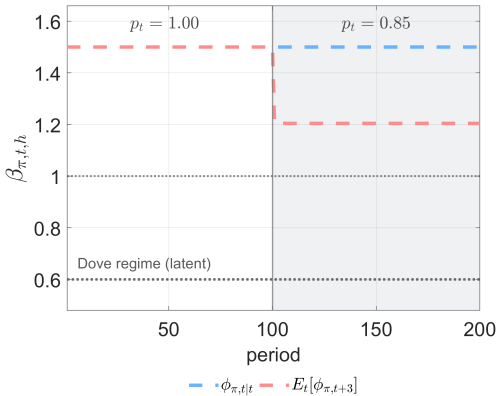


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Survey reports, $t = 1, \ldots, 200$:

	$h=0$		$\cdots$	$h=3$	
t	$\mathbb{E}_t[i_t]$	$\mathbb{E}_t[\pi_t]$		$\mathbb{E}_t[i_{t+3}]$	$\mathbb{E}_t[\pi_{t+3}]$
1	4.50	3.00	$\cdots$	4.35	2.90
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
200	3.00	2.00	$\cdots$	2.40	2.50

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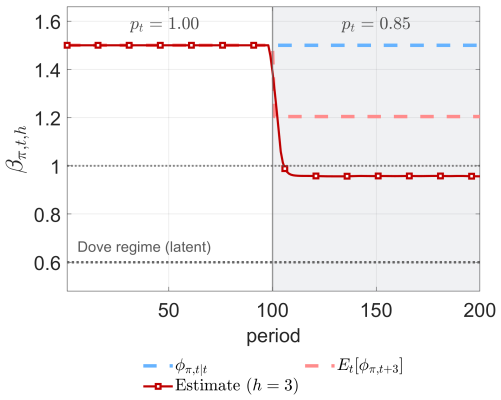
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TVP estimation:

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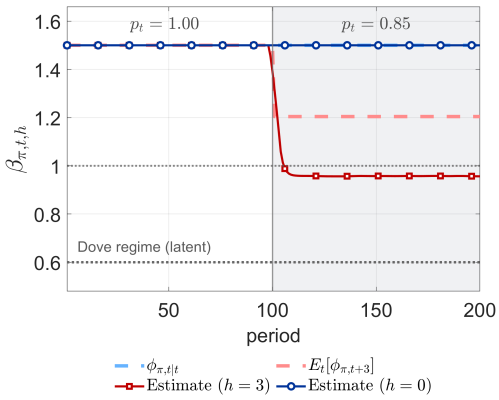
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t	$\mathbb{E}_t[i_t]$	$\mathbb{E}_t[\pi_t]$		$\mathbb{E}_t[i_{t+3}]$	$\mathbb{E}_t[\pi_{t+3}]$
1	4.50	3.00	$\cdots$	4.35	2.90
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
200	3.00	2.00	$\cdots$	2.40	2.50

TVP estimation:

$$\mathbb{E}_t[i_{t+h}] = \beta_{\pi,t,h}\, \mathbb{E}_t[\pi_{t+h}] + e_{t,h}$$

- ✗ Forecast: 0.96

Numerical Illustration - Forecast vs. Nowcast



- Always Hawk: $\phi_{\pi,t|t} = 1.50$.
- But starts to doubt: $\mathbb{E}_t[\phi_{\pi,t+3}] = 1.20$.

Survey reports, $t = 1, \ldots, 200$:

	$h = 0$		$\cdots$	$h = 3$	
t	$\mathbb{E}_t[i_t]$	$\mathbb{E}_t[\pi_t]$		$\mathbb{E}_t[i_{t+3}]$	$\mathbb{E}_t[\pi_{t+3}]$
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$$\mathbb{E}_t[i_{t+h}] = \beta_{\pi,t,h}\, \mathbb{E}_t[\pi_{t+h}] + e_{t,h}$$

- ✗ Forecast: 0.96 ✓ Nowcast: 1.50

Roadmap

1. Simple Model of Perceived vs. Actual Rule
2. Identification of Perceived Rule
3. **Empirical Estimation of Perceived Rule**
4. Subjective Perception Markov Switching New Keynesian Model

Data

Perceived Rule

- *Baseline:* BCEI + SPF, 1983-2024, quarterly.
- *Robustness:* BCFF microdata.

Data

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Subjective model →

Example survey, period t

	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
	nowcast		forecasts		
$i_{t+h\|t}$	5.3	5.1	4.8	4.5	4.2
$\pi_{t+h\|t}$	3.2	2.9	2.7	2.5	2.4
$u_{t+h\|t}$	4.1	4.2	4.3	4.3	4.4

- **Identifying assumption:** survey forecasts reflect the reporter's **subjective model**.
- The monetary block of that model can be represented by a Taylor rule.

Data

Perceived Rule

- *Baseline:* BCEI + SPF, 1983-2024, quarterly.
- *Robustness:* BCFF microdata.

Actual Rule

- *Source:* Greenbook (Tealbook).
- *Sample:* 1969-2019, FOMC frequency.

Subjective model →

Example survey, period t

	$h=0$	$h=1$	$h=2$	$h=3$	$h=4$
	nowcast		forecasts		
$i_{t+h\|t}$	5.3	5.1	4.8	4.5	4.2
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Empirical Framework

Forward-looking Taylor rule specification:

Actual rule (Greenbook) Boivin (2005) and Coibion and Gorodnichenko (2011)

$$i_t = (1 - \rho_t)\left[\phi_{\pi,t}\pi^{GB}_{t+1|t} + \phi_{x,t}\left(-2 \cdot u^{GB}_{t+1|t}\right)\right] + \rho_{1,t} i_{t-1} + \rho_{2,t} i_{t-2} + c_t + \varepsilon_t$$

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Nowcast-based perceived rule ($h = 0$)

$$i_{t|t} = (1-\rho_{t|t})\left[\phi_{\pi,t|t}\pi^{\text{BC/SPF}}_{t+1|t} + \phi_{x,t|t}\left(-2\cdot u^{\text{BC/SPF}}_{t+1|t}\right)\right] + \rho_{1,t|t} i_{t-1|t} + \rho_{2,t|t} i_{t-2|t} + \tilde{c}_t + e_t$$

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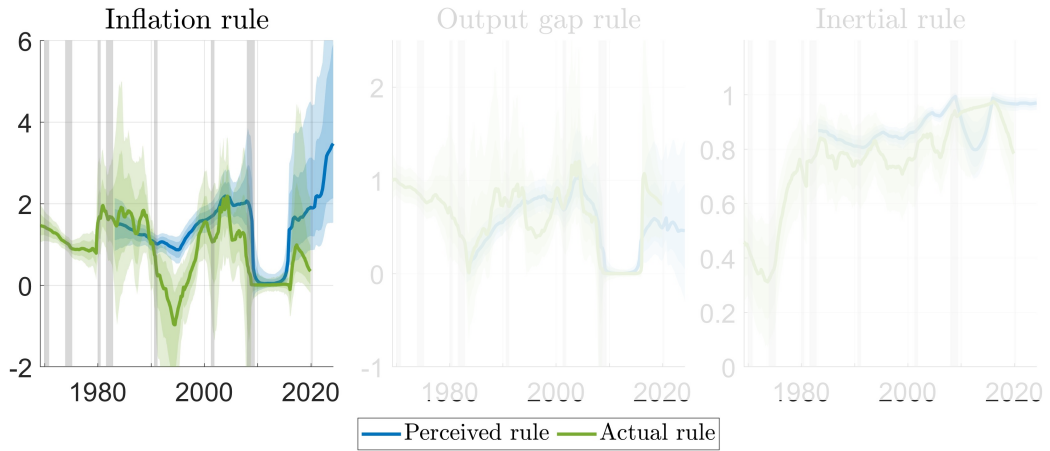
$$i_{t|t} = (1-\rho_{t|t})\left[\phi_{\pi,t|t}\pi^{\text{BC/SPF}}_{t+1|t} + \phi_{x,t|t}\left(-2\cdot u^{\text{BC/SPF}}_{t+1|t}\right)\right] + \rho_{1,t|t} i_{t-1|t} + \rho_{2,t|t} i_{t-2|t} + \tilde{c}_t + e_t$$

Forecast-based perceived rule ($h \geq 1$)

$$i_{t+h|t} = (1-\rho_{t,h})\left[\beta_{\pi,t,h}\pi^{\text{BC/SPF}}_{t+h+1|t} + \beta_{x,t,h}\left(-2\cdot u^{\text{BC/SPF}}_{t+h+1|t}\right)\right] + \rho_{1,t,h} i_{t+h-1|t} + \rho_{2,t,h} i_{t+h-2|t} + \tilde{c}_{t,h} + e_{t,h}$$

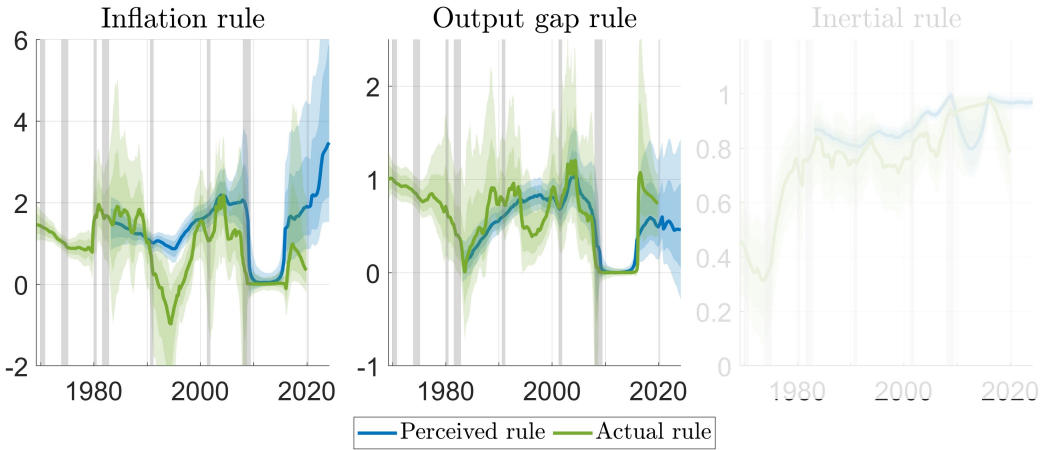
Use Stock and Watson (1998) and Boivin (2005) + stochastic volatility.

Perceived vs. Actual Rule



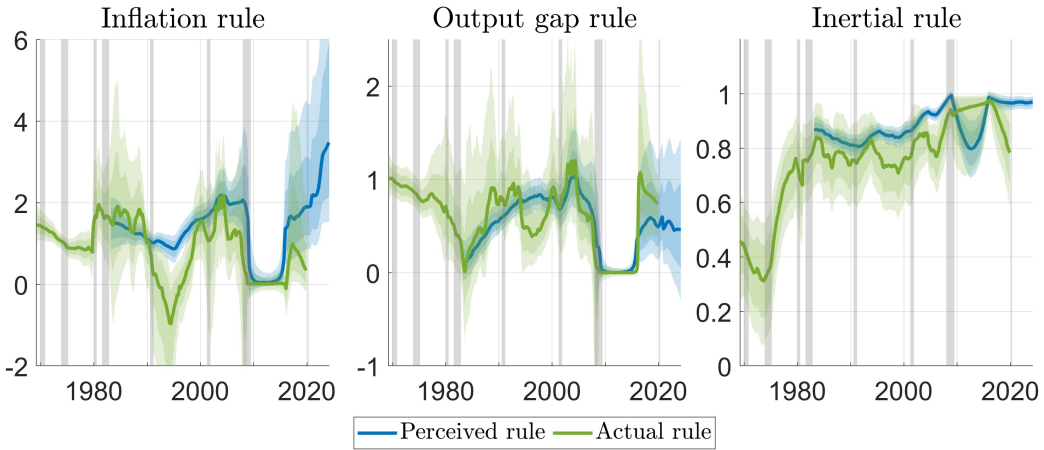
The perceived inflation rule follows the actual rule except for the Risk Management period in the 90s.

Perceived vs. Actual Rule



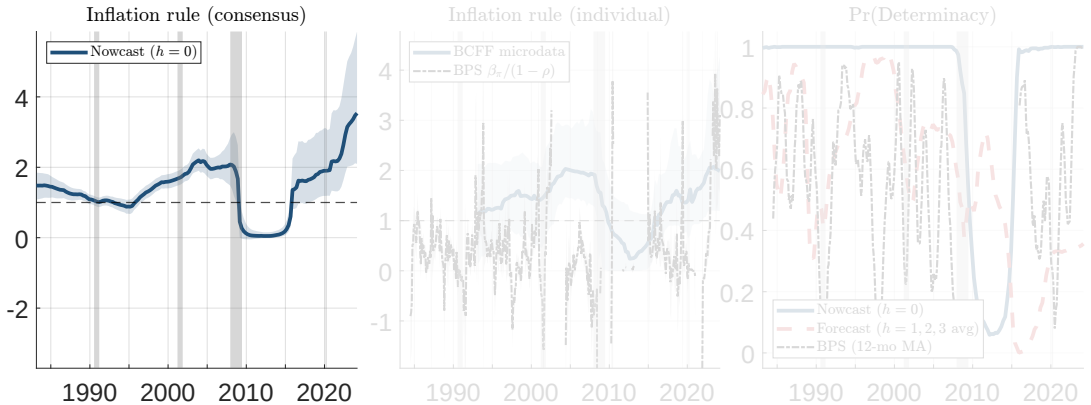
The perceived importance of output gap stabilization increases over time.

Perceived vs. Actual Rule



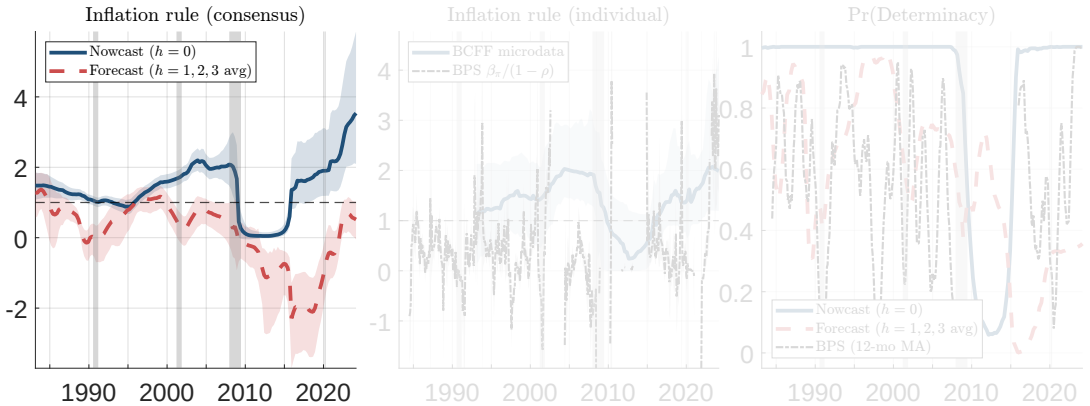
The perceived rule has become more and more inertial over time.

Restoration of Perceived Taylor Principle



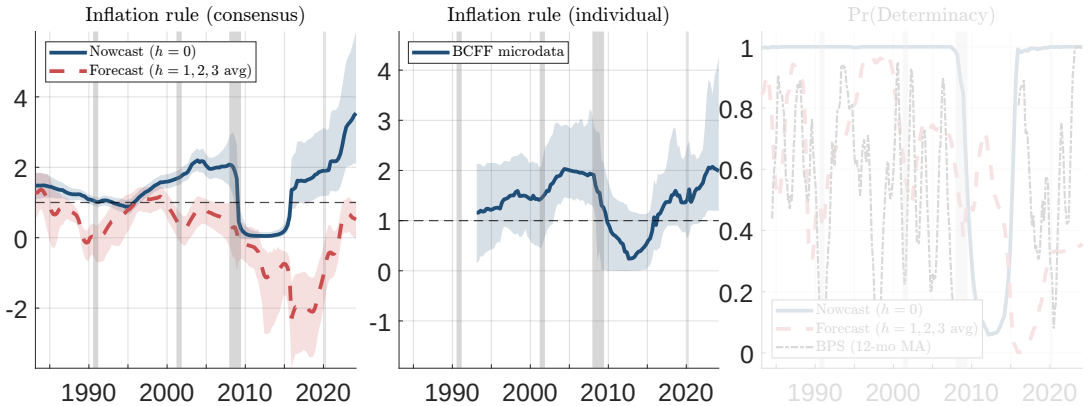
Nowcast-based estimates satisfy the Taylor principle (> 1), but forecast-based estimates do not.

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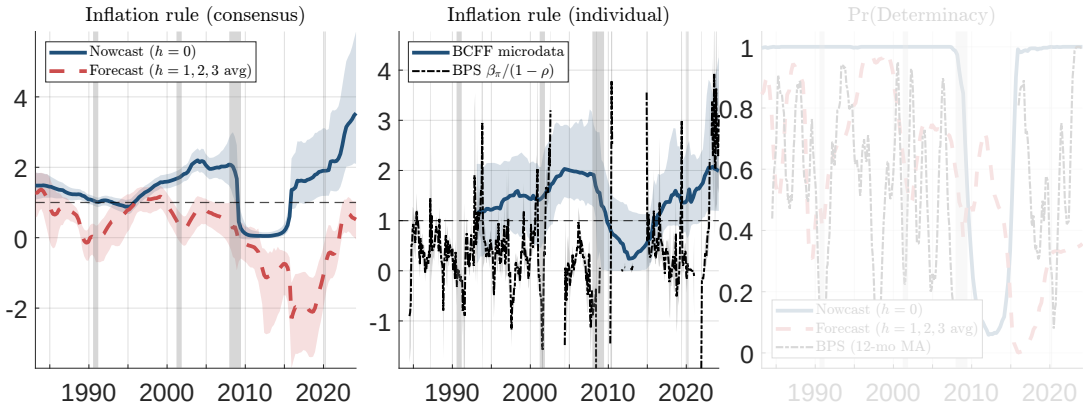
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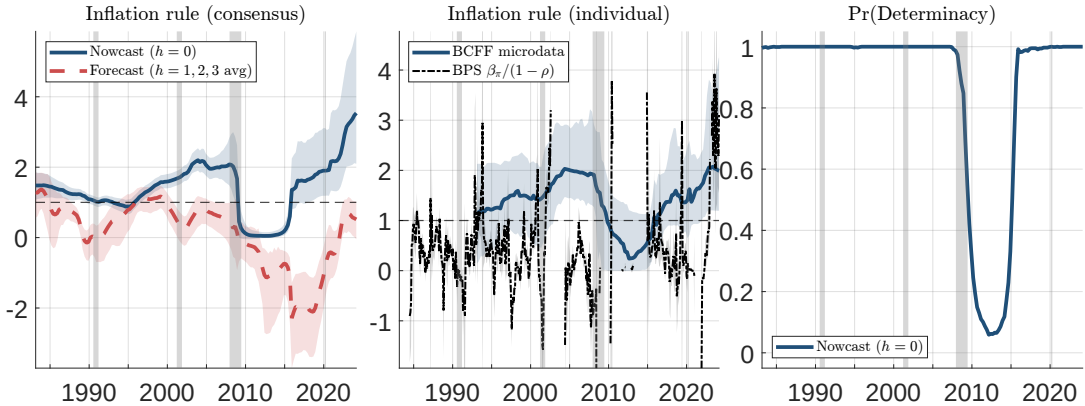
Individual-level estimates are also consistent with the Taylor principle.

Restoration of Perceived Taylor Principle



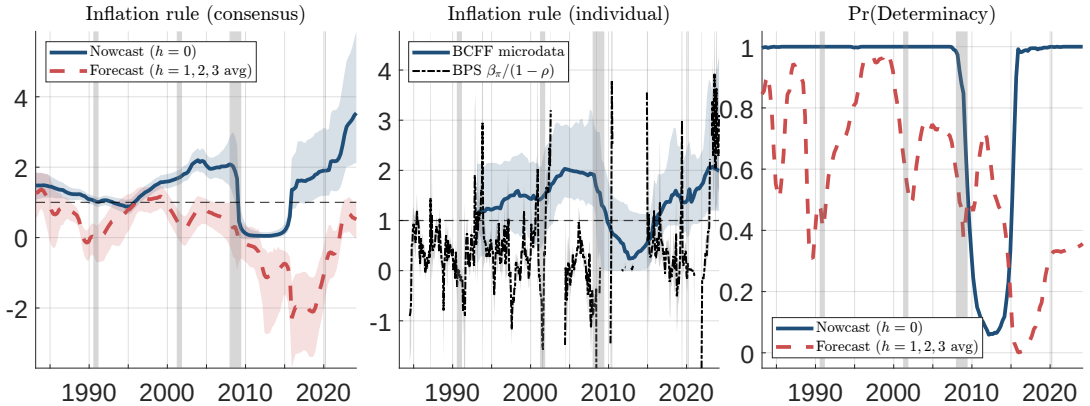
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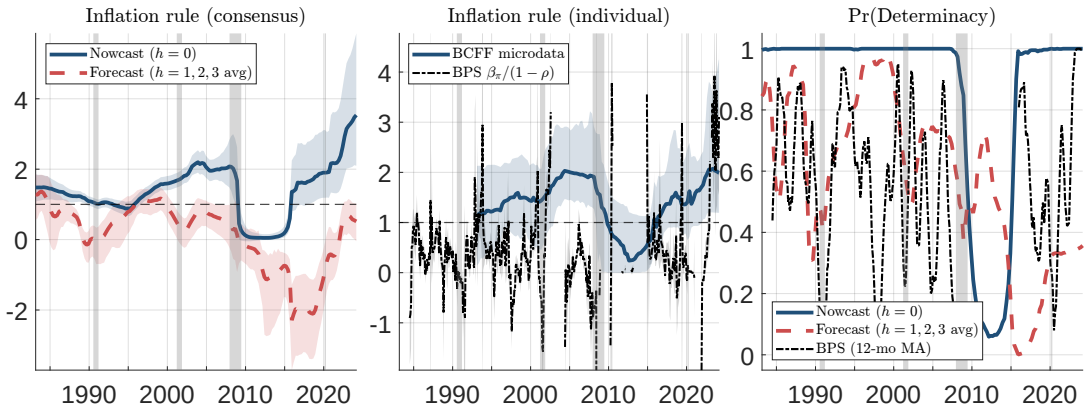
Determinacy evaluated period-by-period through the textbook NK model.

Restoration of Perceived Taylor Principle



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Roadmap

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SP-MSNK Model: Framework and Solution

$$\left.\begin{array}{l}\text{Perceived } s_t^p \in \{1 \cdots S_p\} \text{ under } \textit{subjective } \widehat{\mathbf{P}} \\ \text{Actual } s_t^f \in \{1 \cdots S_f\} \text{ under } \mathbf{P}\end{array}\right\} \Longrightarrow \text{True state } \ell = (s_t^p, s_t^f)$$

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1. **Perceived Law of Motion.** Agents believe, and solve, given $s_t^p = j$:

$$\mathbf{\Gamma}_{0,j}^{\mathrm{P}} E_t^{\mathrm{P}}[\mathbf{y}_{t+1}] + \mathbf{\Gamma}_{1,j}^{\mathrm{P}} \mathbf{y}_t + \mathbf{\Gamma}_{2,j}^{\mathrm{P}} \mathbf{y}_{t-1} + \mathbf{\Gamma}_{3,j}^{\mathrm{P}} \boldsymbol{\varepsilon}_t^p = \mathbf{0} \quad \Longrightarrow \quad \mathbf{y}_{t|t} = \widehat{\mathbf{A}}_j^{\mathrm{PLM}} \mathbf{y}_{t-1} + \widehat{\mathbf{B}}_j^{\mathrm{PLM}} \boldsymbol{\varepsilon}_t^p$$

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2. **Expectations.** From that PLM, for $s_t^p = j$:

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3. **Actual Law of Motion.** Substituting $E_t^{\mathrm{P}}[\cdot]$, the economy obeys, given ℓ:

$$\mathbf{\Gamma}_{0,\ell}^{\mathrm{A}} E_t^{\mathrm{P}}[\mathbf{y}_{t+1}] + \mathbf{\Gamma}_{1,\ell}^{\mathrm{A}} \mathbf{y}_t + \mathbf{\Gamma}_{2,\ell}^{\mathrm{A}} \mathbf{y}_{t-1} + \mathbf{\Gamma}_{3,\ell}^{\mathrm{A}} \boldsymbol{\varepsilon}_t = \mathbf{0} \quad \Longrightarrow \quad \mathbf{y}_t = \mathbf{A}_\ell^{\mathrm{ALM}} \mathbf{y}_{t-1} + \mathbf{B}_\ell^{\mathrm{ALM}} \boldsymbol{\varepsilon}_t$$

Implication 1 – Determinacy depends on perception

Model ($S_p = S_f = 1$)

$$x_t = E_t^{\mathrm{P}}[x_{t+1}] - \tau\left(i_t - E_t^{\mathrm{P}}[\pi_{t+1}] - r_t^n\right)$$

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Rules

Perceived: $i_t = r_t^n + \phi_\pi^{\mathrm{P}} \pi_t$

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Under FIRE $\left(\phi_\pi^{\mathrm{P}} = \phi_\pi^{\mathrm{F}}\right)$

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Peg (Galí 2015), then $\phi_\pi^{\mathrm{F}} > 1$.

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Determinacy is required to attain the target allocation $\pi_t = x_t = 0$.

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Peg (Galí 2015), then $\phi_\pi^{\mathrm{F}} > 1$.

Dropping FIRE ($\phi_\pi^{\mathrm{P}} \neq \phi_\pi^{\mathrm{F}}$)

$$\pi_t = x_t = 0 \;\Rightarrow\; / \Leftarrow \; i_t = r_t^n$$
$$\text{Determinacy} \iff \phi_\pi^{\mathrm{P}} > 1$$

Peg restored, without $\phi_\pi^{\mathrm{F}} > 1$.

Determinacy is required to attain the target allocation $\pi_t = x_t = 0$.

Policy and empirical implications: Not $\phi_\pi^F > 1$, but $\phi_\pi^P > 1$.

SP-MSNK Model: Setup

IS and Phillips curve

$$x_t = E_t^{\mathrm{P}}\left[x_{t+1}\right] - \tau\left(i_t - E_t^{\mathrm{P}}\left[\pi_{t+1}\right]\right) + z_{Dt}$$
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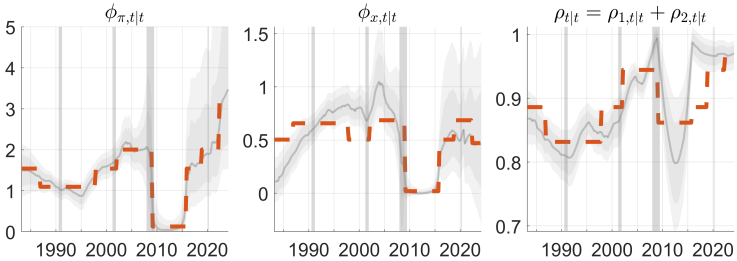
Rules ($S_p = S_f > 1$)

$$\text{Perceived: } i_t = \left(1 - \rho_{s_t^p}\right)\left[\phi_{\pi,s_t^p} E_t^{\mathrm{P}}[\pi_{t+1}] + \phi_{x,s_t^p} E_t^{\mathrm{P}}[x_{t+1}]\right] + \rho_{1,s_t^p} i_{t-1} + \rho_{2,s_t^p} i_{t-2}$$

$$\text{Actual: } i_t = \left(1 - \rho^{\mathrm{F}}_{s_t^f}\right)\left[\phi^{\mathrm{F}}_{\pi,s_t^f} E_t^{\mathrm{P}}[\pi_{t+1}] + \phi^{\mathrm{F}}_{x,s_t^f} E_t^{\mathrm{P}}[x_{t+1}]\right] + \rho^{\mathrm{F}}_{1,s_t^f} i_{t-1} + \rho^{\mathrm{F}}_{2,s_t^f} i_{t-2} + \varepsilon_{Mt}$$

- s_t^p evolves under $\widehat{\mathbf{P}}$, s_t^f under $\mathbf{P}$, with $\mathcal{S}^p = \mathcal{S}^f = \mathcal{S} = \{1, \ldots, S\}$.
- The Fed responds to E_t^{P}: to isolate the role of perceived vs. actual rules.

Calibration



- Baseline: $(s_t^p) = (s_t^f)$, giving ℓ.
- Dove: counterfactuals only.
  Lubik and Schorfheide 2004, Farmer, Waggoner, and Zha 2011

Regime-specific rule

	ϕ_π	ϕ_x	ρ	$\widehat{P}_{kk}$
1	3.18	0.47	0.97	0.79
2	2.00	0.69	0.94	0.86
3	1.54	0.50	0.89	0.84
4	1.09	0.66	0.83	0.87
5	0.13	0.02	0.86	0.84
6: Dove	0.77	0.68	0.60	0.98

Structural (Pfäuti 2023)

$$\beta = 0.99 \quad \tau = 1.00 \quad \kappa = 0.057$$
$$\rho_D = \rho_S = 0.80$$

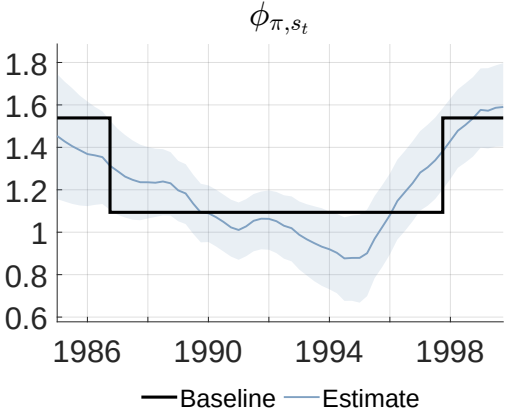
ϕ_x is annualized. $\rho = \rho_1 + \rho_2$ is total inertia.

Implication 2 – Source of the Great Moderation

Consensus: the Great Moderation followed a shift to a hawkish policy rule.
Clarida, Gali, and Gertler (2000), Coibion and Gorodnichenko (2011), and Bianchi (2013)

Implication 2 – Source of the Great Moderation

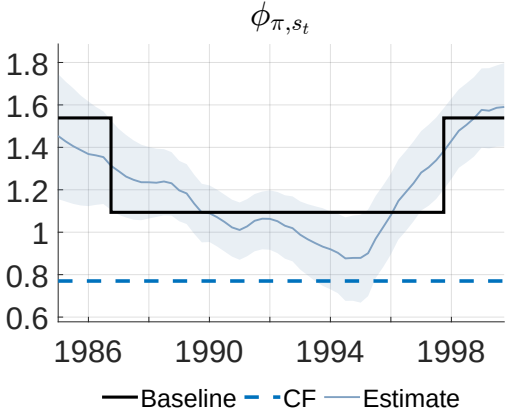
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Baseline: Hawk **perception**, Hawk actual.

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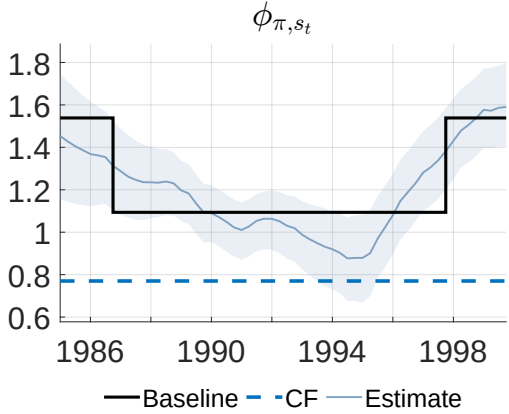


Baseline: Hawk **perception**, Hawk actual.

1. Hawk **perception**, **Dove** actual
   $\Rightarrow$ *Direct policy channel*

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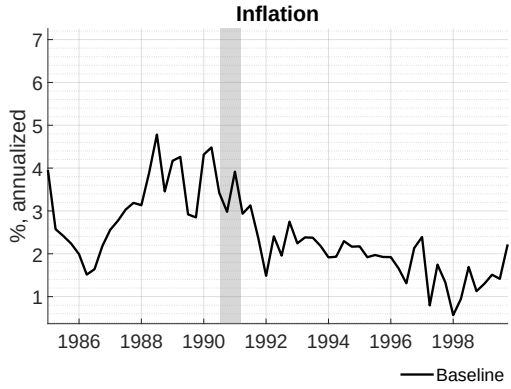
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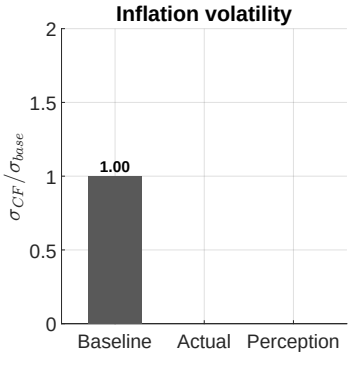
1. Hawk **perception**, **Dove** actual
   ⇒ *Direct policy channel*

2. **Dove** **perception**, Hawk actual
   ⇒ ***Perception*** *channel*

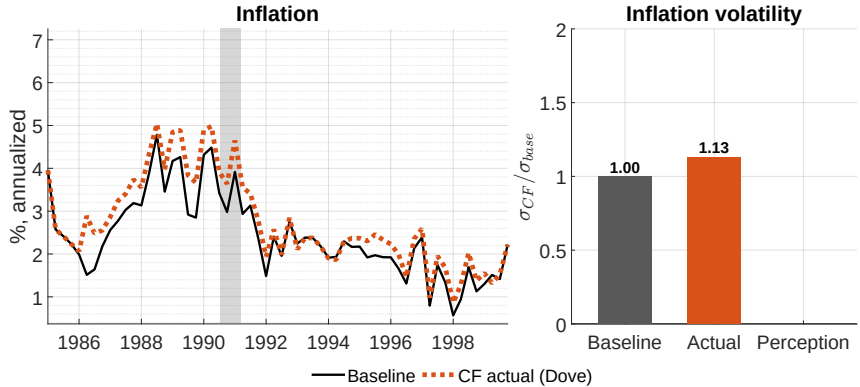
Q. What was the role of perception?

Implication 2 – Source of the Great Moderation

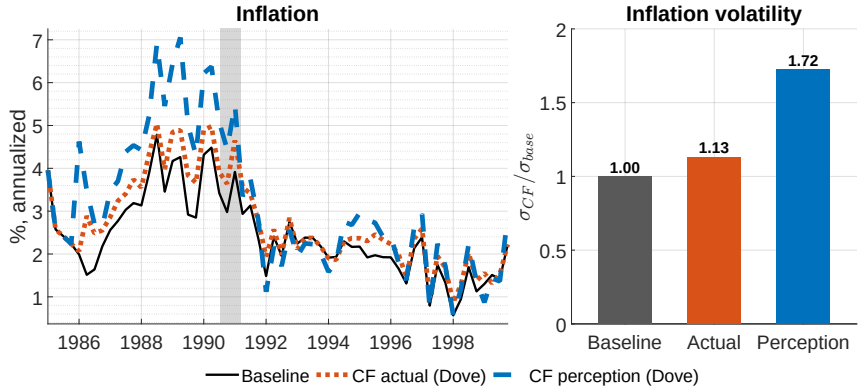




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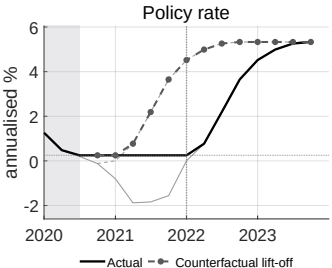


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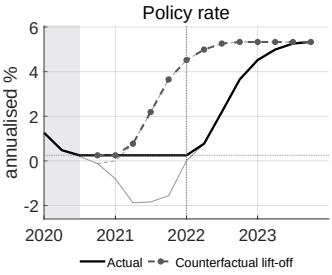


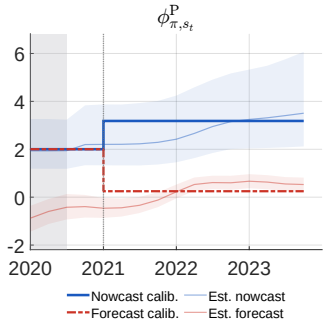
- Dovish beliefs amplify the level and volatility more than the dovish rule.

Implication 3 – Misreading Perception (2021–2023)



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Counterfactual: lift-off pulled at **2021Q1**.

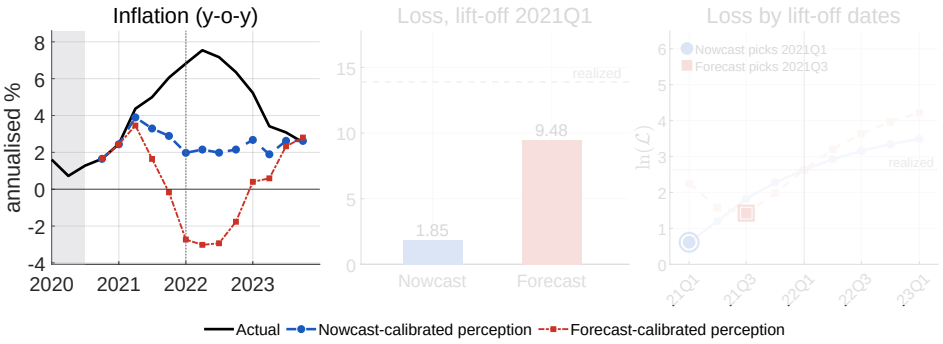
Two central banks read the *same* path:

1. **Nowcast**-calibrated ϕ_{π,s_t^p}
2. **Forecast**-calibrated ϕ_{π,s_t^p}

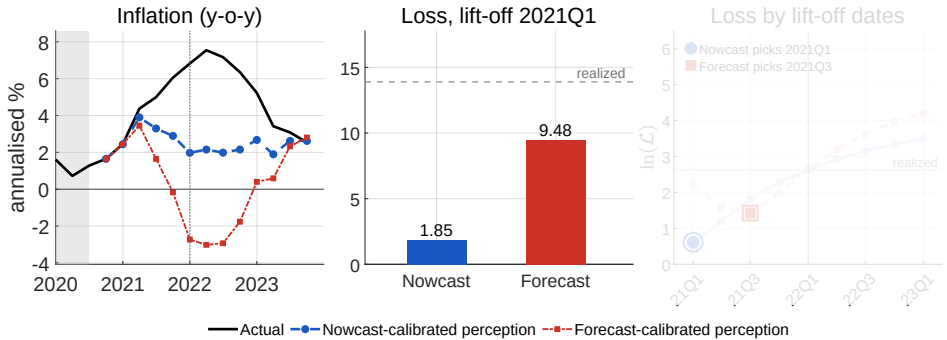
Each scores loss with ($\lambda = 0.007$):

$$\mathcal{L} = \mathbb{E}\left[\left(\pi_t^{\text{yoy}} - 2\right)^2 + \lambda x_t^2\right]$$

Implication 3 – Cost of misreading perception

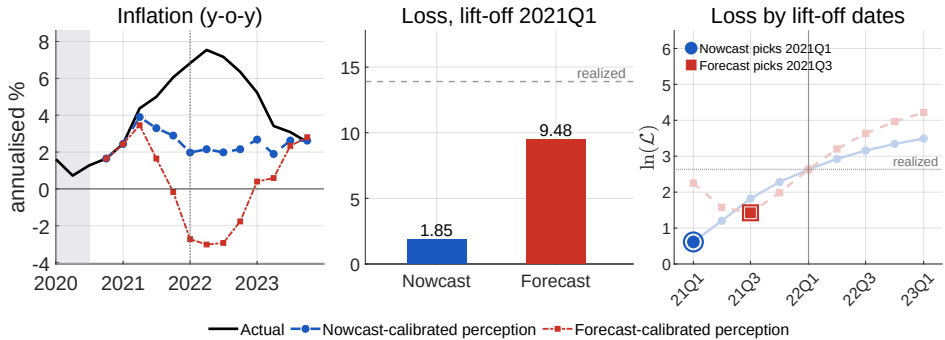


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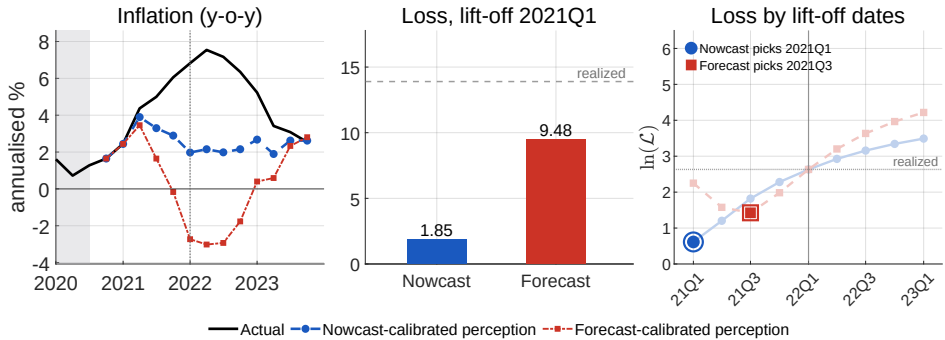
- The same policy path is scored differently under the two **perceptions**.

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- The forecast-based Fed rates the nowcast-optimal lift-off worse than its own optimal lift-off.

Implication 3 – Cost of misreading perception



- The same policy path is scored differently under the two **perceptions**.
- The forecast-based Fed rates the nowcast-optimal lift-off worse than its own optimal lift-off.
- Misreading **perception** leads to miscalculating policy outcomes.

Conclusion and Contributions

$$\underbrace{\text{“Is the Fed hawkish?”}}_{\text{Actual Rule } (\phi_{\pi,t})} < \underbrace{\text{“Do people } \textit{perceive} \text{ the Fed as hawkish?”}}_{\text{Perceived Rule } (E_t[\phi_{\pi,t}] \equiv \phi_{\pi,t|t})}$$

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   - Determinacy depends on the **perceived** Taylor principle, not the actual one.

Conclusion and Contributions

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     (The Great Moderation, post-pandemic lift-off)

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2. Interest rate forecasts provide a biased estimate of the true **perception**.